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The Use of Background Features, Template Synthesis and Deep Neural Networks in Document Forgery Detection

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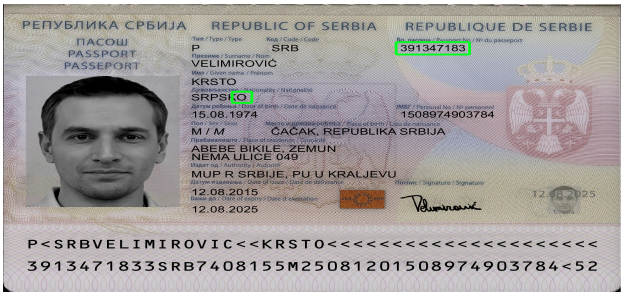
***Abstract*—Document manipulation is a recently arising problem, especially with the rapid spread of fabrication technology. The tools to alter documents are now publicly available and can result in high quality forgeries, indistinguishable from genuine ones. Forged documents may wreak havoc on many processes dependent on the validity of the document, leading to lasting consequences such as financial loss. Therefore, the process of identifying a document that has been altered is essential. A system that is capable of scrutinizing documents as either forged or genuine through discriminative features (such as distortions or character misalignment) can assist industries with heavily reliance on documents for processes such as identity verification. Most of the documents involved in such processes have sufficiently complex backgrounds. We present a computer-vision-based system that detects changes in the background of the aforementioned documents as a result of manipulations made to its contents through the use of image subtraction. The system takes an image as input and then classifies the document as genuine or forged. Our proposed system produces an accuracy of 95% using CNN on unaligned images as well as 100% for aligned images.**

***Index Terms*—Document, forgery, detection, genuine, CNN, background, subtraction, forged, splicing, copy-move**

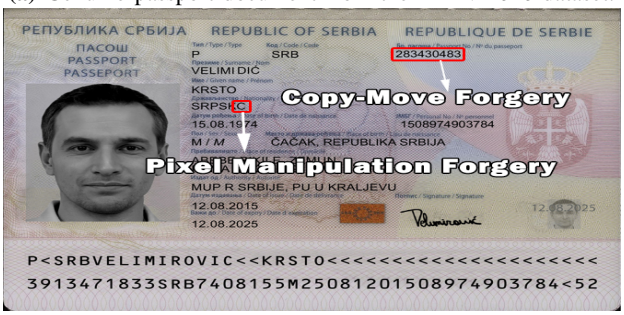
I. Introduction

Forgery is the process in which an item in the form of a document is altered or created to resemble a genuine counterpart in an attempt to deceive others. Forgeries manifest from modifications made to existing documents. Determining whether a document is forged or genuine (see Fig. 1) is crucial for most verification procedures [1], [2], which is used in most business aspects and can cause potential data breaches if a forged document was used as a genuine document. The issue regarding digital documents unlike the physical documents that can be inspected closely is that they can be easily modified with publicly available tooling such as photo editors as well as scanners and printers [3].

For such types of documents, the text, including the surrounding areas can be valuable indicators of potential forgery. Specifically, these forms of text-based forgery are detectable by analyzing variations in the rotation and the alignment of characters [4].



(a) Genuine passport document from the MIDV-2020 dataset.



(b) Forgery afflicted with Pixel manipulation and copy-move.

Fig. 1: Genuine and forged documents

Various methods of textual forgeries exist but are often classified into the following types: Pixel Manipulation involves altering pixels on the document to modify characters existing on the document. Splicing is the process in which a group of pixels taken, often from another document. This is intended to cascade a character by laying another over it. Splicing produces forgeries that may be easily spotted due to the inconsistency of its photo-characteristic properties (such as color, noise, and font size) with its surroundings. Nonetheless,

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Copy-Move forgeries, which are a form of splicing in which the source of the spliced pixels is taken from the same image; providing relatively more consistent properties [5]. Fig. 2. depicts the previously mentioned methods.

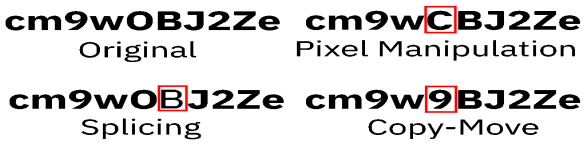


Fig. 2: Methods of Textual forgery

The techniques used to detect forged documents has been often partitioned into 2 classes. Active Forgery Detection is the technique of instilling authenticatable information within an image or document [6]. This information manifests in the form of serial numbers, embedded holograms, watermarks, or other indicators. These forms of authentication are easily verifiable by the issuing body or creator, but are uncommon and often reserved for specific forms of documents. An independent body or party may struggle with verifying these details, as they are still prone to modification and often require specialized equipment to be verified.

In contrast, passive forgery detection is the process of detecting the authenticity of the source image without relying on embedded data.

Forged images may exhibit evidence of modification despite resembling its genuine counterpart (For an example, see Fig. 1a and Fig. 1b). Sharma et. al [6] gives the example of distortion, artifacts and loss of information as symptoms or clues to a possibly forged document. As depicted in Fig. 1b. , it can be observed that in most documents with sufficiently detailed backgrounds, the background of the copied content does not match the patterns of the surrounding background in the event of forgery (see Fig. 3). Sharma et. al [6] further categorizes passive forgery detection of images into multiple classes.



Fig. 3: Examples of background disturbance in forged documents.

As described by Amidi et al. [7], The success of deep-learning based models is attributed to the massive amount of data available for training purposes, however, many deep-learning models can be properly trained with sufficient sizes of data sets through the use of data augmentation, in which the input image is rotated, flipped, recolored, etc. In particular, Convolutional Neural Networks (CNN) are a subclass of neural networks that have shown to be particularly successful for use in image recognition tasks and classification due to proficiency in detecting spatial correlations or features.

The main contribution of this paper is to investigate the use of the aforementioned technique (as shown in Fig. 4.), and evaluate its efficacy using classical machine learning models as well as a convolutional model in the context of the detection of forged documents, measuring the accuracy of the system using the MIDV-2020 [8] extended with manually created forged images. The proposed system is a form of pixel based forgery detection, as the change in intensity of background pixels upon manipulation is indicative of a forgery, as is therefore detected in that manner. However, this form of document manipulation detection has multiple challenges. Namely, the ability to retrieve the backdrop of an image of a blank document without foreground details (such as personal information) in order to exclusively detect manipulation in the background of the document. Furthermore, the classification of unaligned images is challenging as it affects intricate features of the background, therefore a genuine document may be misclassified due to artifacts left during registration and vice versa.

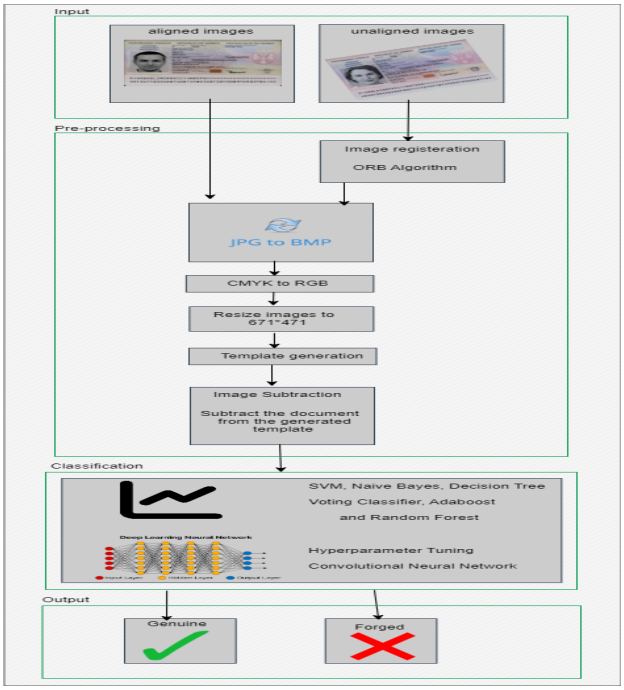


Fig. 4: System overview

II. Related Work

Research involved in such fields investigate common forgery types that can cause the most danger. Several existing approaches were used to detect such forgeries through the use of techniques such as optical character recognition (OCR), statistical models and deep learning models.

A. Classical Approaches

James et al. [4] implemented a random forest based classifier utilizing OCR (using Tesseract [9]) graph features of

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each character in a document such as width, height, and vertical displacement relative to its neighbouring characters on the same line. The proposed system targeted plain digital documents which lack colors or complex backgrounds. Imperfections in the previously mentioned features indicate areas of manipulation. The classifier was capable of reaching 98.57%. Moreover, The authors proposed using pixel-based features from images along with the OCR graph features in order to further enhance performance. Osorio et al. [5] proposed an algorithm that could identify the type of forgery afflicted on the image using a hybrid model consisting of SIFT and Markov processing. The algorithm achieved an accuracy of 91%. Ryu et al. [3] proposed a Support Vector Machine (SVM) based technique which achieved an overall classification accuracy of 80%. The system aimed to detect printed forged documents through the inspection of image qualities, along with the identification of the printer (laser, ink-jet, etc) that was used to produce a forged document.

B. Deep Learning Based Approaches

Marra et al. [10] proposed a conceptual framework for the detection of forged images, with special emphasis on modifications which are not visually obvious. The proposed system consisted of a CNN which automatically extracts defining features from an entire image without resorting to resizing, which is often the cause of loss in detail. The system was evaluated on a data-set representative of real forgeries. Additionally, Fahmi & Sari [11] investigated the use of CNNs of similar architectures for detection of tampered images. Specifically, the models investigated the effect of an additional convolutional layer on the pre-existing two layers on accuracy of the model. The models were evaluated against various data-sets consisting of images afflicted with copy-move or splicing. The most effective model, equipped with an additional convolutional layer, perform with 83% accuracy on average. However, the model proposed was suspected to have overfit. The authors suggested that it could be remedied with a larger data set with a diverse set of forgeries. Zhao et al. [2] developed a neural network capable of editing documents with textual information surrounded by complex backgrounds. The system is capable of disguising the artifacts produced by the modification such as mismatching colors or distortion. Tarek et al. [1] performed a comparative analysis on the accuracy of detecting forged signatures across multiple CNN models (such as ResNet50, VGG-16, etc). The results showed that ResNet-50 was the highest performing model, producing an accuracy of 82.3%. Bunk et al. [12] proposed a system using Patch Classification and LSTM to classify and localize regions in images that have been tampered. The system has achieved an accuracy of 94.86%.

III. Methodology

A. Data Collection

1) Genuine Documents: Due to the scarcity of publicly available documents, we make use of synthetic documents accurately resembling their real counterparts. Notably, we make use of the MIDV-2020 [8] data set. the MIDV-2020 is an extension of the MIDV-500 [13] document data-set and consists of 10 document types, each containing 100 genuine documents (See Fig. 1a). The difference being the increase in diversity of the input fields, faces, and signatures. In addition to another 100 genuine documents for each document type provided with the data-set that were unaligned under the name ”scan_rotated” (See Fig. 5a). Additionally, a photograph, video clip and scanned version of each document was also provided, depicting realistic scenarios where the input is provided from an unaligned image or video frame, this is further discussed in the pre-processing section.

TABLE I: Gap analysis of related work.

Ref	Input Type	Author	Classification Model	Dataset	Accuracy
[4]	Documents	James et. al.	OCR Random Forest	Proprietary	98.57%
[3]	Documents	Ryu et al.	SVM	Proprietary	80%
[5]	Images	Osorio et al.	SIFT Markov Processing	Columbia LAB, CASIA V1	91%
[10]	Images	Marra et al.	CNN	Modified FAU, DSO-1, Korus, NC and MFC.	82.4%
[11]	Images	Fahmi & Sari	CNN	Proprietary, CASIA ITDE and FAU Image Manipulation	83%
	Document	Proposed System	CNN	Modified MIDV-2020	95%

2) Forged Documents: The MIDV-2020 data-set does not include forged documents. Therefore, forged documents were created manually with approximately 14 documents per type (See Fig. 1b), forming 132 forged documents in total for experiment 1. 105 unaligned forged documents were also manually created and registered (as shown in Fig. 4) to conduct experiment 2. Forming up to 237 manually created forged documents to conduct both experiment 1 and 2.



(a) Sample of experiment 2 documents before registration



(b) Sample of experiment 2 documents after registration

Fig. 5: Sample of experiment 2 documents before and after registration.

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B. Deep-Learning Model

The Convolutional Neural Network (CNN) was developed using the Keras API for TensorFlow [14] and consists of several layers, which are specified in Fig. 6. A data augmentation layer was used to make random horizontal flips and random rotations to decrease the rate of over-fitting. Second, a re-scaling layer which re-scales the images to the fraction of 1/255 of their size.

The images are then passed into multiple convolutional layers each followed by a normalization layer. We applied hyperparameter tuning in order to determine the best dropout value for the model to minimize over-fitting while maintaining adequate results. Hyperparameter tuning is the process in which a search space consisting of all possible values constructed for the set of hyperparameters (such as dropout rate, kernel size, etc.). The search space is then examined through random selection or through a bracket style of elimination to obtain values for each hyperparameter in the model. The Tuner [15] is responsible for assessing the quality of the model during the search for more optimal parameter, after a specified number of epochs, the best model is selected and its values for hyperparameters are retained. The model was re-trained after finding optimal hyperparameters on a machine equipped with 16 GB of RAM, a NVIDIA RTX 3070 TI, and an Intel Core i9 10900F. The set of forged documents were manually created with varying levels of detail and quality. A set of genuine photos were moved into the forged data set before being manually altered. Each forged document contains one or more occurrences of pixel manipulation, copy-move and splicing. Similar to the genuine counterparts, each image was subtracted from a generated template to expose background details.

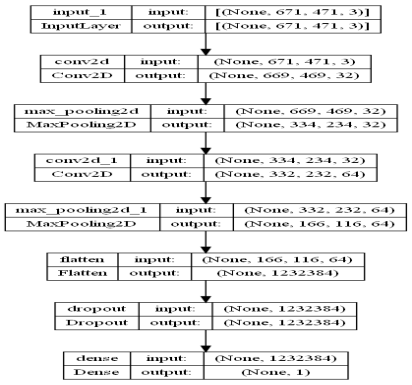


Fig. 6: Architecture of the Convolutional Neural Network

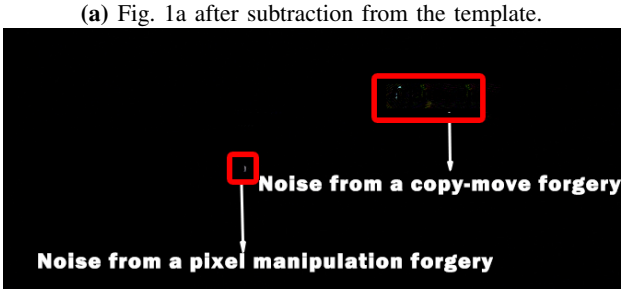
C. Preprocessing

The images were converted to bitmap images to avoid non-deterministic behavior produced by compression or specifics concerning a particular file format, and resized to dimensions of 671x471 as well as converted to RGB from other color formats (such as CMYK) for consistency.

The images are aligned via Image Registration. (See Fig. 4.) Each image was aligned using normalization types such as L1, L2, Hamming, and other forms available in OpenCV's implementation of ORB [16], feature matching algorithm which relies on descriptive key points common to both the source and companion image to compute a transformation capable of producing a candidate for the aligned image from the unaligned source image. The most suitable aligned image is measured through the absolute difference between each aligned candidate and the pre-aligned template for each document type. The candidate with the minimum distance to the template is then selected. Although the absolute difference seems to be a simple heuristic, it has proven to be effective, requiring little manual selection or discarding of unreasonably aligned documents. As previously stated, the manipulated areas of a forged document cause subtle changes to the surrounding background. To further reveal the difference, the input image is subtracted from an empty document (As shown in Fig. 4), containing the genuine background to reveal the difference. This accentuates the area that has been tampered with (See Fig. 7b.) whereas a genuine document would produce one as seen in Fig. 7a. In the event that a template for the background is unavailable, a sufficiently large amount of sample documents could be used to synthesize an approximate template. Assuming that the document consists of dark test with a lighter background, pixels representing the background can be extracted by taking the brightest pixel from the samples, forming an image such as the one shown in Fig. 8. Although the generated template contains artifacts (e.g the photograph, or common text), it is sufficient for its purpose.



(a) Fig. 1a after subtraction from the template.



(b) Fig. 1b after subtraction from the template

Fig. 7: The result of the subtraction of a genuine and forged from the template.

Once a template has been generated, the data-set is generated by subtracting each image by it's template using saturating arithmetic, meaning that resulting values are clamped to

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the range 0 to 255 rather than wrapping semantics produced by underflow or overflow (See Eq. 1. for a formal definition). This cascades artifacts arising from direct subtraction of pixel values.

$$h(x, y) = \max\{0,\ \min\{255,\ f(x, y)\ -\ g(x, y)\}\} \qquad (1)$$

IV. Experiments and Result Discussion

Both experiments follow similar setups. First, a template was generated for a sample of (aligned) genuine documents, the sample size used for both experiments was 100 images. After preprocessing the images according to the requirements of the experiment (see Fig. 4.), the images are assigned to a training and test set for each type of document after subtraction from the template, which produces the input to the models. Each model was evaluated on each type of document model and their respective accuracy is measured.

A. Experiment 1

The first experiment aims to assess the accuracy of a selection of models on the post-subtracted images produced from perfectly aligned images of different types of documents.

1) Setup: The first experiment uses aligned images to test the aforementioned technique, which was tested on 8 different types of documents, each containing 132 forged documents and 162 genuine documents. All models were trained on the same data set. The models consist of the proposed convolutional neural network (see Fig. 6, and 7 statistical models listed in Table III. The aforementioned models were selected to resemble the current state of the art models for binary classifications as well as selected from the related work.

2) Results: Table II shows the results produced by each model in each type of document present. Due to the stark contrast between the noise levels of a genuine and post-subtraction forged image, approaches such as the CNN are highly effective where as classical algorithms such as SVM and Naive Bayes perform adequately. Other classical models have varying performance, which is justified in the remainder of the section. Regardless of any potential correlations amidst the features, the Naive Bayes classifier considers each of these features to contribute independently to the confidence. The single Decision Tree classifier outperforms a Random Forest model in terms of average accuracy. This is likely due to the environment of the experiment, as well as the variance produced by the differing noise patterns in the forged images, causing it to underfit. The voting classifier consists of an ensemble of the classical models, wherein the final classification result is produced from the majority vote of the sub-classifiers. However the decision is unweighted, therefore many incorrect classification results produced by sub-classifiers will dominate the outcome over less votes for the correct result produced for a particular class.

B. Experiment 2

1) Setup: Unaligned scanned copies of documents (obtained from MIDV-2020 [8]) were then first aligned, followed by template generation using the same procedure described and used in Experiment 1. A subset of the aligned documents produced in the previous step was used to generate the template, the remainder was then subtracted from the template (as shown in Fig. 4.) and used to train the models as the training and validation set.



Fig. 8: Template document of srb_passport from MIDV-2020

2) Results: As shown in Table III, overall accuracy produced from experiment 2 has regressed as expected, due to the noise generated by the alignment errors produced during registration. Notably, the results demonstrate the weak generalizability of the Decision Tree. The Random Forest demonstrated higher accuracy, likely due to the increase in variation in the data-set overall. This variation has caused the Naive Bayes to be biased towards producing a classification result of forged, even for genuine documents due to the alignment noise. Although CNN demonstrated a lower accuracy than SVM in the second experiment, CNN is likely more desirable than SVM for this particular use case as it produces significantly lower false positives (forged documents classified as genuine) than SVM as shown in the confusion matrices (generated using [17]) for both classifiers (see Fig. 9.)

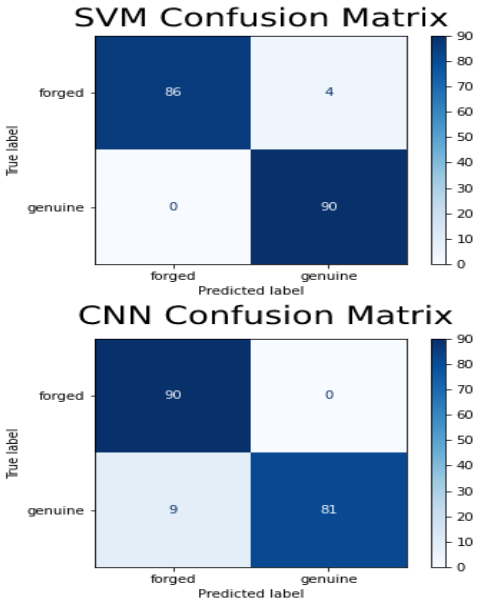


Fig. 9: Confusion Matrix evaluated on unaligned documents

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TABLE II: Accuracy of each document class from Experiment 1 comparing selected classical machine learning algorithms with the CNN model.

	CNN	***SVM***	***Naive Bayes***	***Decision Tree***	***Voting Classifier***	***AdaBoost***	***Random Forest***
alb_id	*100.00%*	*100.00%*	*100.00%*	*90.91%*	*93.94%*	*90.91%*	*87.88%*
aze_passport	*100.00%*	*100.00%*	*100.00%*	*57.58%*	*75.76%*	*57.58%*	*54.55%*
est_id	*100.00%*	*100.00%*	*94.12%*	*55.88%*	*52.94%*	*52.94%*	*52.94%*
fin_id	*100.00%*	*96.97%*	*96.97%*	*93.94%*	*63.64%*	*78.79%*	*54.55%*
grc_passport	*100.00%*	*100.00%*	*100.00%*	*100.00%*	*54.55%*	*100.00%*	*54.55%*
lva_passport	*100.00%*	*100.00%*	*100.00%*	*54.55%*	*100.00%*	*57.58%*	*100.00%*
rus_internalpassport	*100.00%*	*100.00%*	*100.00%*	*56.25%*	*56.25%*	*56.25%*	*56.25%*
svk_id	*100.00%*	*100.00%*	*92.86%*	*75.00%*	*75.00%*	*71.43%*	*67.86%*
Average accuracy	***100.00%***	***99.62%***	***97.99%***	***73.01%***	***71.51%***	***70.69%***	***66.07%***

TABLE III: Accuracy of Experiment 2 comparing selected classical machine learning algorithms with the CNN model

Classifier	**CNN**	**SVM**	**Naive Bayes**	**Decision Tree**	**Voting Classifier**	**AdaBoost**	**Random Forest**
Accuracy	**95.00%**	**97.78%**	**75.00%**	**55.56%**	**76.11%**	**57.78%**	**71.11%**

V. Conclusion and Future Work

In conclusion, computer vision has the potential to revolutionize the field of document forgery detection by providing a reasonably reliable and automated way to analyze and verify the authenticity of a document. Using machine learning, models can take advantage of various background features of a document and compare them to known patterns and characteristics of genuine documents. Two experiments were conducted; the first experiment processed aligned documents taken from an artificially sterile environment where no lighting conditions or environmental noise were introduced and taken into account. The second experiment was conducted on unaligned documents to measure the efficiency of the proposed system in real environments where alignment and environment noise exist. The MIDV-2020 data set was used for both experiments; forged documents manually created by mimicking various types of forgeries. Using a set of genuine documents for each type of document, a generic template was generated, and then images were subtracted from the template to construct the genuine and forged data set. The post-subtraction images were then fed to one CNN model and six classical machine learning models. The most optimal and consistent models in both experiments were CNN and SVM, scoring 100.00% and 96.62%, respectively, in experiment 1 and 95.00% and 97.78% in experiment 2. Despite the fact that SVM had higher results than CNN, CNN is more reliable in this particular use case as it produces significantly lower false positives than SVM which is very sensitive in such field. Future work includes extending the forged document data set with more diverse types of forgeries with more environment noise such as different lighting conditions or the documents being contorted. The proposed system can be combined with an OCR-based approach to further improve accuracy.

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